

CE 206 (18 Batch) Final Quiz Solve

Set- A

1. You are designing a spherical tank (Figure-1) to hold water for a small village in a developing country. 10

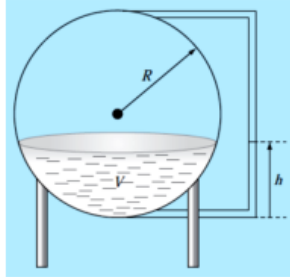


Figure-1

The volume of liquid it can hold can be computed as:

$$V = \pi h^2 \frac{[3R - h]}{3}$$

where V = volume [m^3], h = depth of water in tank [m], and R = the tank radius [m]. If $R = 3$ m, to what depth must the tank be filled so that it holds 30 m^3 ? Use **Regula-Falsi method** and **Newton Raphson method** (initial guess 3) for maximum three iterations.

Answer

Regula Falsi Method

```
>> [root,fx,ea,iter]=regulafalsi(@(h)h^2-h^3/9-10/pi,0.1,3,0.0001,3)

root =

    2.0243

fx =

   -0.0070

ea =

    1.6160

iter =

     3
```

Newton Raphson Method

```
>> [root,ea,iter]=newtraph(@(h)h^2-h^3/9-10/pi,@(h)(2*h-(h^2)/3),3,0.0001,3)

root =

    2.0269

ea =

    0.0067

iter =

     3
```

2. The table below provides the year wise registered number of motorcycles according to BRTA in Dhaka city.

10

Year	2005	2007	2009	2011	2013	2015	2017
Number	13,362	14,520	24,675	35,195	28,086	46,758	75,251

- Using not-a-knot cubic interpolation determine the number of motorcycles registered in the year 2012.
- Plot the motorcycle growth data using “interp1” function along with “spline”. Show proper axis title and legend.
- Use “polyfit” and “polyval” function to predict the number of motorcycles in year 2018 using a 6th order polynomial.

Answer:

a.

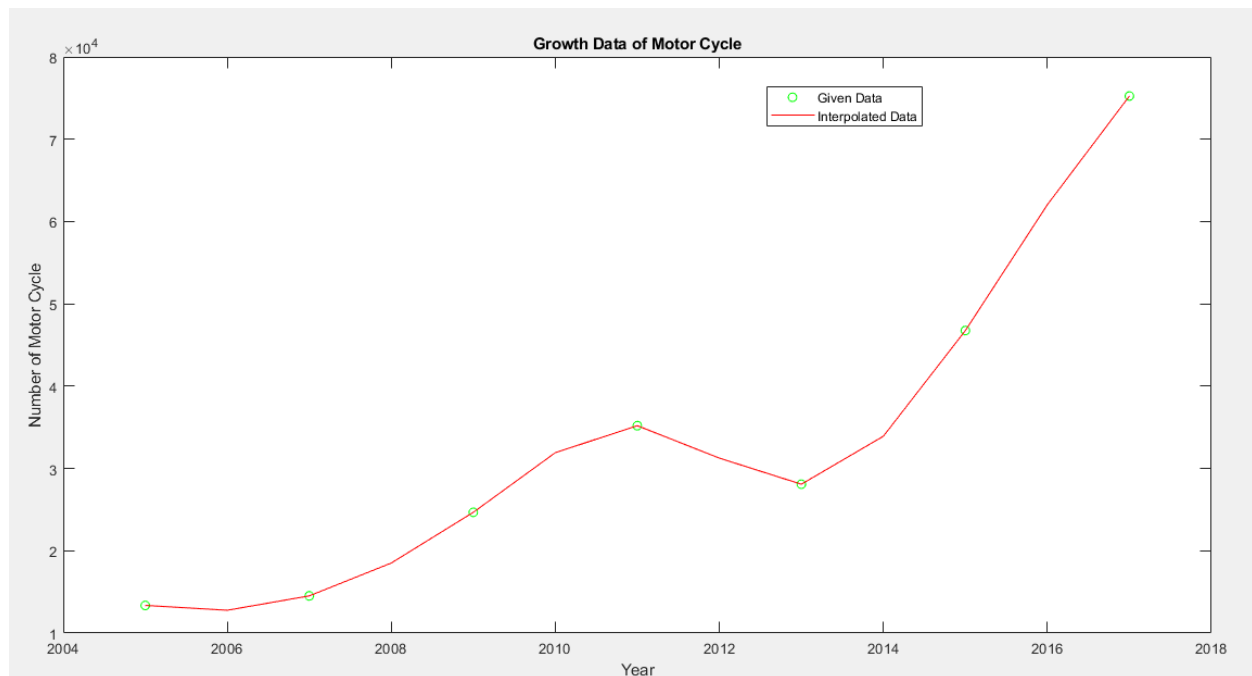
```
>> Year=[2005 2007 2009 2011 2013 2015 2017];
>> Number=[13362 14520 24675 35195 28086 46758 75251];
>> yy=spline(Year,Number,2012)

yy =

    3.1256e+04
```

b.

```
>> Year=[2005 2007 2009 2011 2013 2015 2017];
>> Number=[13362 14520 24675 35195 28086 46758 75251];
>> yy=2005:2017;
>> nn=interp1(Year,Number,yy,'spline');
>> plot(Year,Number,'go',yy,nn,'r-');
>> legend('Given Data','Interpolated Data');
>> xlabel('Year');
>> ylabel('Number of Motor Cycle');
>> title('Growth Data of Motor Cycle');
>>
```



C.

```
>> p = polyfit(Year,Number,6)
Warning: Polynomial is badly conditioned. Add points with distinct X values,
the polynomial, or try centering and scaling as described in HELP POLYFIT.
> In polyfit (line 84)

p =

    1.0e+19 *
    -0.0000    0.0000   -0.0000    0.0000   -0.0000    0.0042   -1.4056

>> y=polyval(p,2018)

y =

    57344
```

3. Solve the following 2D truss (Figure-2) using LU factorization. Find the values of F_{AB} , F_{BC} , F_{CD} , F_{DA} , F_{BD} , A_y , A_x and C_y . Mention if the member forces are in tension or compression. Report all the values in a neat tabular format. 10
- X = Last three Digits of your student ID (for example: ID= 1504001, X = 001)**

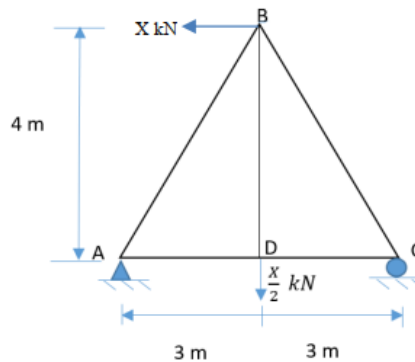


Figure-2

Answer:

```
>> A=zeros(8)
```

$$A =$$
[illegible][illegible]

A									
8x8 double									
	1	2	3	4	5	6	7	8	9
1	0.6000	0	0	1	0	0	1	0	
2	0.8000	0	0	0	0	1	0	0	
3	-0.6000	0.6000	0	0	0	0	0	0	
4	-0.8000	-0.8000	0	0	-1	0	0	0	
5	0	-0.6000	-1	0	0	0	0	0	
6	0	0.8000	0	0	0	0	0	1	
7	0	0	1	-1	0	0	0	0	
8	0	0	0	0	1	0	0	0	

A =

```

0.6000    0    0    1.0000    0    0    1.0000    0
0.8000    0    0    0    0    1.0000    0    0
-0.6000   0.6000    0    0    0    0    0    0
-0.8000  -0.8000    0    0   -1.0000    0    0    0
0   -0.6000  -1.0000    0    0    0    0    0
0   0.8000    0    0    0    0    0    1.0000
0    0    1.0000  -1.0000    0    0    0    0
0    0    0    0    1.0000    0    0    0

```

```

>> B=[0 0 32 0 0 0 0 16]';
>> [L,U]=lu(A);
>> d=L\B;
>> x=U\d;

```

x	
8x1 double	
	1
1	-36.6667
2	16.6667
3	-10
4	-10.0000
5	16
6	29.3333
7	32
8	-13.3333

4. The basic differential equation of the elastic curve for a simply supported beam having concentrated load at mid span is given as: 10

$$EI \frac{d^2y}{dx^2} = \frac{Px}{2} \quad (0 \leq x \leq \frac{L}{2})$$

Where, E = the modulus of elasticity, L = Beam Span and I = the moment of inertia.

The boundary conditions are $y(0) = y(L) = 0$, $y(L/2) = \frac{PL^3}{48EI}$

Solve for the deflection of the beam up to mid span using the shooting method and plot a graph.

Use the following values:

$E = (100 + \text{Last three digits of your Student ID}) \text{ GPa}$,

$I = 30,000 \text{ cm}^4$,

$L = 5 \text{ m}$

$P = 50 \text{ KN}$

Compare the numerical results with the analytical solution in the same graph.

Use different line styles, colors, shapes, and appropriate legends for plotting.

N.B. Analytical Solution:

$$y = \frac{Px}{12EI} \left(\frac{3L^2}{4} - x^2 \right) \quad (0 \leq x \leq \frac{L}{2})$$

Answer:

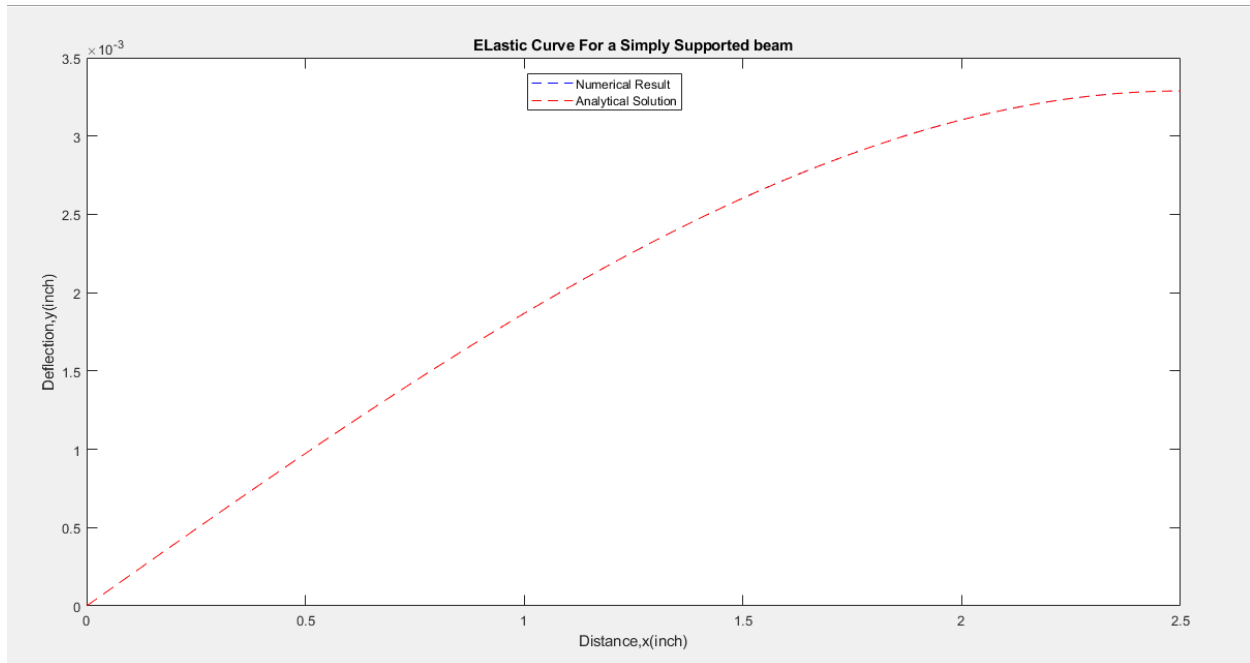
```
beamdeflect.m  res.m  func.m  +
1 function s=beamdeflect(X,Y)
2 E=132*10^9;I=3*10^-4;L=5;P=50*1000;
3 s=[Y(2);-(P*X)/(2*E*I)];
4 end
```

```
beamdeflect.m  res.m  func.m  +
1 function r = res(X)
2 E=132*10^9;I=3*10^-4;L=5;P=50*1000;
3 [x,y]=ode45(@beamdeflect, [0 2.5], [0 X]);
4 r=y(length(x),1)-((P*L^3)/(48*E*I));
5 end
```

```
beamdeflect.m  res.m  func.m  +
1 function [value] = func(x)
2 E=132*10^9;I=3*10^-4;L=5;P=50*1000;
3 value= (P.*x)/(12*E*I).*((3*L^2)/4-x.^2);
4 end
```

```
>> z=0:0.025:2.5;
>> hold off
>> plot(z,func(z),'b--');
>> hold on;
>> [x,y]=ode45(@beamdeflect,[0 2.5],[0 fzero(@res,0)]);
>> plot(x,y(:,1),'r--');
```

```
>> xlabel('Distance,x(inch)');  
>> ylabel('Deflection,y(inch)');  
>> legend('Numerical Result','Analytical Solution');  
>> title('ELastic Curve For a Simply Supported beam');
```



Numerical Result is same as analytical solution. There was a mistake in the question at the differential equation.

5. The horizontal surface area A_s (m^2) of a lake at a particular depth can be computed from volume by differentiation: 10

$$A_s(z) = \frac{dV(z)}{dz}$$

where V = volume (m^3) and z = depth (m) as measured from the surface down to the bottom.

The average concentration of a substance $\bar{c}$ (g/m^3) in a lake where the area $A_s(m^2)$ varies with depth $z(m)$ can be computed by integration:

$$\bar{c} = \frac{\int_0^Z c(z) A_s(z) dz}{\int_0^Z A_s(z) dz}$$

where Z = the total depth (m). Determine the area of the sections at depth $z(m)$ and average concentration based on the following data:

z, m	0	4	8	12	16
$V, 10^6 m^3$	9.8175	5.1051	1.9635	0.3927	0.0000
$C, g/m^3$	10.2	8.5	7.4	5.2	4.1

Answer:

```
>> z=0:4:16;
>> v=[9.8175 5.1051 1.9635 0.3927 0]*10^6;
>> C=[10.2 8.5 7.4 5.2 4.1];
>> A=gradient(v,4);
>> E=trapz(z,C.*A);
>> F=trapz(z,A);
>> C_avg=E/F;
>> C_avg
```

```
C_avg =

    8.2252
```